ZHANHAO HU

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EDUCATION

Tsinghua University, Beijing *2017 - 2023*
Ph.D., Computer Science and Technology
Advisor: Bo Zhang and Xiaolin Hu
Dissertation: The Practicality of Physical Adversarial Examples for Deep Learning Models.
Research: adversarial robustness, privacy, and brain-inspired algorithms in deep learning.

Tsinghua University, Beijing *2013 - 2017*
B.S., Mathematics and Physics
Advisor: Xiaolin Hu
Dissertation: STDP-based Learning for Spiking Neural Networks

RESEARCH INTERESTS

My research develops principled methods for building and red-teaming robust and secure AI systems, spanning large language models (LLMs), computer vision, agentic systems, and multimodal and embodied AI. My recent interests include LLM alignment, prompt injection defenses, secure agentic system design, and adversarial robustness in multimodal and embodied AI systems.

RESEARCH EXPERIENCE

University of California, Berkeley January 2024 - Present
Postdoctoral Researcher *Berkeley, California*

- Advised by Prof. David Wagner, affiliated with the Department of Electrical Engineering and Computer Science (EECS) and the Institute for Data Science (BIDS)
- Research focuses on LLM security and safety, including model-level and agentic system-level attacks and defenses.
- Led 6+ projects on red-teaming, detection, and defense across threat models in LLM and embodied systems.
- Collaborated with students and faculty on LLM security, agentic system security, and multimodal robustness.

Tsinghua University July 2023 - December 2023
Researcher *Beijing, China*

- Worked with Prof. Xiaolin Hu at the Tsinghua Laboratory of Brain and Intelligence (THBI).
- Research focused on physical adversarial robustness for computer vision models.

PUBLICATIONS

(Sorted by time; * for equal contribution)

Under review* & *Preprint

1. **Zhanhao Hu**, Dennis Jacob, Xiao Huang, Zhaorun Chen, Bo Li, and David Wagner (2026) Twin Agent: Context Residual Compression for Privilege Separated Agents (preprint)
2. **Zhanhao Hu***, Xiao Huang*, Patrick Mendoza, Emad A. Alghamdi, Basel Alomair, Raluca Ada Popa, and David Wagner (2026). GradShield: Alignment Preserving Finetuning (under review)
3. Zi Yin, Peilin Chai, Siyuan Huang, **Zhanhao Hu** (2026). Test-time Adversarial Takeover: A Real-time Hijacking Interface against Robotic Diffusion Policies (under review)
4. Jesson Wang, **Zhanhao Hu** (2026). Safety Offset Learning: When Benign Fine-Tuning Misaligns LLMs (under review)
5. Dennis Jacob, Emad Alghamdi*, **Zhanhao Hu***, Basel Alomair, and David Wagner (2025). Better Privilege Separation for Agents by Restricting Data Types (under review)
6. Wei Zhang, **Zhanhao Hu**, Xiao Li, Xiaopei Zhu, and Xiaolin Hu (2025). A Single Set of Adversarial Clothes Breaks Multiple Defense Methods in the Physical World (preprint)
7. Qiongxiu Li, Lixia Luo, Agnese Gini, Changlong Ji, **Zhanhao Hu**, Xiao Li, Chengfang Fang, Jie Shi, Xiaolin Hu (2024). Perfect Gradient Inversion in Federated Learning: A New Paradigm from the Hidden Subset Sum Problem (preprint)

Published

8. Jesson Wang*, **Zhanhao Hu***, and David Wagner (2026). JULI: Jailbreak Large Language Models by Self-Introspection (ICLR).
9. Xiaopei Zhu, Guanning Zeng, **Zhanhao Hu**, Jun Zhu, and Xiaolin Hu (2026). Physical Adversarial Clothing Evades Visible-Thermal Detectors via Non-Overlapping RGB-T Pattern (CVPR)
10. Julien Piet, Xiao Huang, Dennis Jacob, Annabella Chow, Maha Alrashed, Geng Zhao, **Zhanhao Hu**, Chawin Sitawarin, Basel Alomair, and David Wagner (2025). Jailbreaksovertime: Detecting jailbreak attacks under distribution shift (AISec)
11. Dennis Jacob, Hend Alzahrani, **Zhanhao Hu**, Basel Alomair, and David Wagner (2025). Prompt-Shield: Deployable Detection for Prompt Injection Attacks (CODASPY)
12. Xiaopei Zhu, Siyuan Huang, **Zhanhao Hu**, Jianmin Li, Jun Zhu, Xiaolin Hu (2025). Physical Adversarial Examples for Person Detectors in Thermal Images Based on 3D Modeling. (TPAMI)
13. **Zhanhao Hu**, Julien Piet, Geng Zhao, Jiantao Jiao, and David Wagner (2024). Toxicity Detection for Free. Advances in Neural Information Processing Systems (NeurIPS Spotlight)
14. Zhi Cheng, **Zhanhao Hu**, Yuqiu Liu, Jianmin Li, Hang Su, Xiaolin Hu (2024). Full-Distance Evasion of Pedestrian Detectors in the Physical World. Advances in Neural Information Processing Systems (NeurIPS)
15. Xiao Li, Wei Zhang, Yining Liu, **Zhanhao Hu**, Bo Zhang, Xiaolin Hu (2024). Language-Driven Anchors for Zero-Shot Adversarial Robustness. In Proceedings of the IEEE/CVF Conference on Computer

Vision and Pattern Recognition (CVPR).

16. Xiao Li, Qiongxiu Li, **Zhanhao Hu**, Xiaolin Hu (2024). On the Privacy Effect of Data Enhancement via the Lens of Memorization. IEEE Transactions on Information Forensics and Security (IEEE TIFS).

17. Xiaopei Zhu, Yuqiu Liu, **Zhanhao Hu**, Jianmin Li, and Xiaolin Hu (2024) Infrared Adversarial Car Stickers. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

18. Xiaopei Zhu, **Zhanhao Hu**, Siyuan Huang, Jianmin Li, Xiaolin Hu (2023). Hiding from Infrared Detectors in Real World with Adversarial Clothes. Applied Intelligence.

19. **Zhanhao Hu**, Wenda Chu, Xiaopei Zhu, Hui Zhang, Bo Zhang, Xiaolin Hu (2023). Physically Realizable Natural-Looking Clothing Textures Evade Person Detectors via 3D Modeling. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

20. Tong Wang, Xiaohui Kuang, Qianjin Du, **Zhanhao Hu**, Huan Deng, Gang Zhao. (2023). Driving into Danger: Adversarial Patch Attack on End-To-End Autonomous Driving Systems Using Deep Learning. In 2023 IEEE Symposium on Computers and Communications (ISCC)

21. **Zhanhao Hu**, Siyuan Huang, Xiaopei Zhu, Fuchun Sun, Bo Zhang, Xiaolin Hu (2022). Adversarial texture for fooling person detectors in the physical world. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR Oral).

22. Xiaopei Zhu, **Zhanhao Hu**, Siyuan Huang, Jianmin Li, Xiaolin Hu (2022). Infrared invisible clothing: Hiding from infrared detectors at multiple angles in real world. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR Oral).

23. **Zhanhao Hu**, Jun Zhu, Bo Zhang, Xiaolin Hu (2022). Amplification Trojan network: Attack deep neural networks by amplifying their inherent weakness. Neurocomputing, 505, 142-153.

24. **Zhanhao Hu**, Tao Wang, Xiaolin Hu (2017). An STDP-based supervised learning algorithm for spiking neural networks. In Neural Information Processing: 24th International Conference (ICONIP).

AWARDS & HONORS

1. NeurIPS Spotlight (top 2%), "Toxicity Detection for Free," 2024
2. CVPR Oral Presentation (top 4%), "Adversarial Texture for Fooling Person Detectors," 2022
3. CVPR Oral Presentation (top 4%), "Infrared Invisible Clothing," 2022
4. Recognized as Outstanding Graduate of the Computer Science Department, 2023

TEACHING & MENTORSHIP

- *Teaching Assistant*, Neural and Cognitive Computation (No.80240642), Tsinghua University, Autumn 2018 & Autumn 2019. — Assisted with tutorials and grading; Delivered a guest lecture on deep learning tooling (PyTorch/TensorFlow).
- *Mentor*, UC Berkeley, 2024–Present — Mentored 3+ undergraduate students in LLM security and safety research.
- *Mentor*, Tsinghua University, 2017–2023 — Mentored 5+ undergraduate and junior Ph.D. students in adversarial example research.

PROFESSIONAL SERVICE

Reviewer

- Journal Reviewer: TIP, TPAMI, TNNLS, PR
- Conference Reviewer: NeurIPS, ICML, CVPR, ICLR, ICCV, ECCV, AAAI, EMNLP, ICIST, ICACI, ICICIP, ISNN

PATENTS

1. Xiaolin Hu, **Zhanhao Hu**, Wenda Chu, Xiaopei Zhu, Three-dimensional modeling method and apparatus for flexible object, electronic device, and storage medium, WO2024250729A1, 2024-12-12
2. Xiaolin Hu, **Zhanhao Hu**, Wenda Chu, Xiaopei Zhu, Method and apparatus for generating camouflage pattern, electronic device, and storage medium, WO2024234714A1, 2024-11-21
3. Xiaolin Hu, Xiaopei Zhu, **Zhanhao Hu**, Siyuan Huang, Jianmin Li, Manufacturing method and using method of infrared camouflage pattern and infrared camouflage garment, CN114557507A, 2022-05-31
4. Xiaolin Hu, Xiaopei Zhu, **Zhanhao Hu**, Siyuan Huang, Jianmin Li, Countermeasure image generation method and apparatus, and target cover processing method, CN114550217A, 2022-05-27
5. Xiaolin Hu, **Zhanhao Hu**, Xiaopei Zhu, Siyuan Huang, Target detector safety testing method and device, electronic device and storage medium, CN113780280B, 2021-12-10
6. Xiaolin Hu, **Zhanhao Hu**, Xiaopei Zhu, Siyuan Huang, Target detector safety testing method and device, electronic device and storage medium, CN113780433B, 2021-12-10